

Class activity-10.07.2026

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1.Student Performance :

```
install.packages("ggplot2")
library(ggplot2)
student <- data.frame(
  Student_ID = c(101,102,103,104,105,106,107,108,109,110,
    111,112,113,114,115,116,117,118,119,120),
  Attendance = c(95,88,76,90,82,98,74,91,85,79,
    93,87,96,81,89,77,94,83,92,80),
  Internal_Marks = c(92,85,70,89,80,96,68,90,82,75,
    91,84,95,78,86,72,93,81,90,76),
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE",
    "CSE","AI","IT","ECE","CSE","AI","IT",
    "ECE","CSE","AI","IT","ECE")
)
print(student)
str(student)ggplot(student,
  aes(x = Attendance,
    y = Internal_Marks,
    color = Department)) +
  geom_point(size = 4) +
  labs(
```

```

title = "Student Academic Performance Dashboard",

x = "Attendance (%)",

y = "Internal Marks",

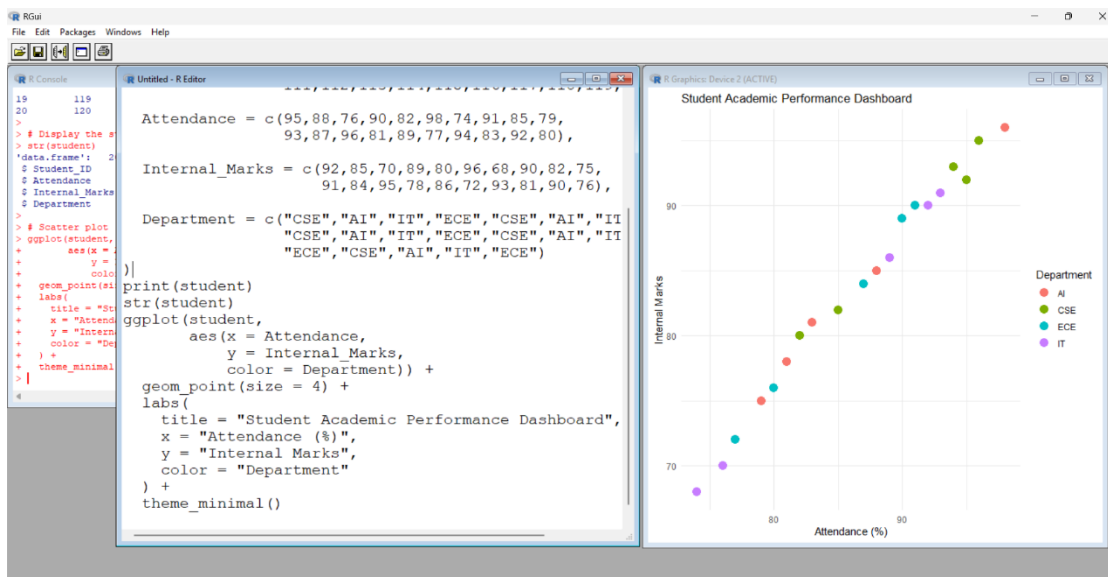
color = "Department"

) +

theme_minimal()

```

Output :



2.Electric Vehicle Market Analysis :

Code:

```

install.packages("ggplot2")

library(ggplot2)

ev <- data.frame(

  Vehicle_Model = c("Nexon EV", "Tiago EV", "MG ZS EV", "BYD Atto 3",
    "Tata Curvv EV", "Mahindra XUV400", "Hyundai Kona",
    "BYD Seal", "Kia EV6", "MG Windsor EV"),

  Battery_Capacity = c(30, 24, 50, 60, 55, 40, 39, 82, 77, 45),

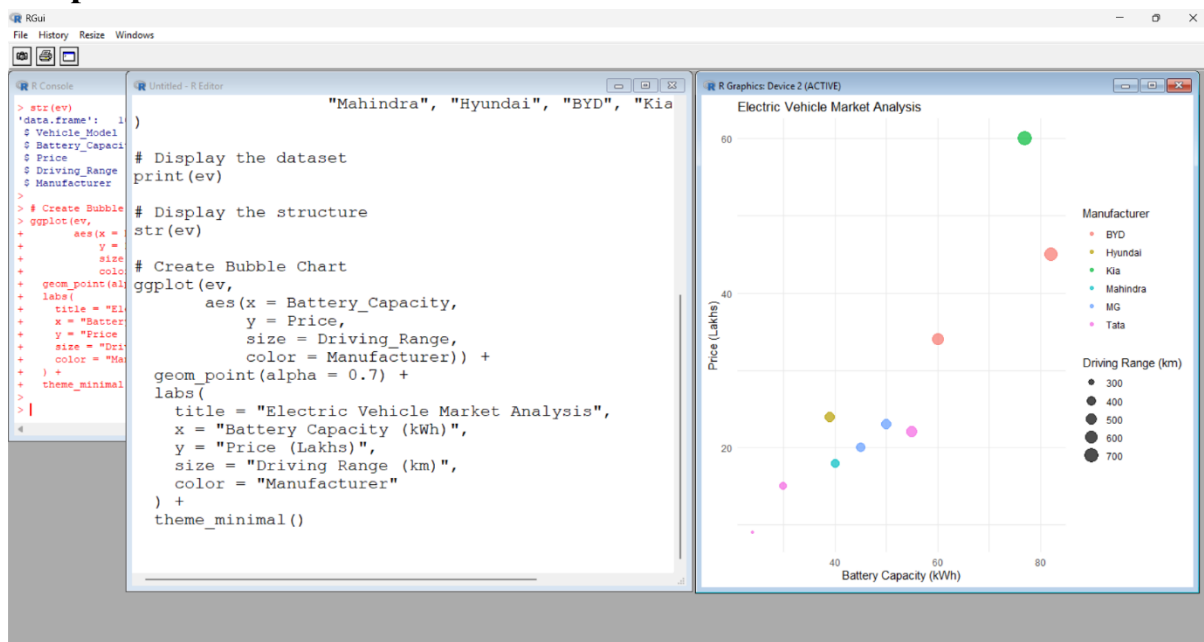
```

```

Price = c(15, 9, 23, 34, 22, 18, 24, 45, 60, 20),
Driving_Range = c(325, 250, 461, 521, 500, 375, 452, 650, 708, 420),
Manufacturer = c("Tata", "Tata", "MG", "BYD", "Tata",
                 "Mahindra", "Hyundai", "BYD", "Kia", "MG")
)
print(ev)
str(ev)
ggplot(ev,
       aes(x = Battery_Capacity,
           y = Price,
           size = Driving_Range,
           color = Manufacturer)) +
geom_point(alpha = 0.7) +
labs(
  title = "Electric Vehicle Market Analysis",
  x = "Battery Capacity (kWh)",
  y = "Price (Lakhs)",
  size = "Driving Range (km)",
  color = "Manufacturer"
) +
theme_minimal()

```

Output :



3.Hospital Patient Admission Analysis :

Code :

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
hospital <- data.frame(
```

```
  Department =
```

```
  c("Cardiology","Neurology","Orthopedics","Pediatrics","General"),
```

```
  Number_of_Patients = c(120,85,95,140,180),
```

```
  Average_Treatment_Cost = c(50000,65000,40000,30000,25000)
```

```
)
```

```
print(hospital)
```

```
str(hospital)
```

```
ggplot(hospital,
```

```
  aes(x=Department,
```

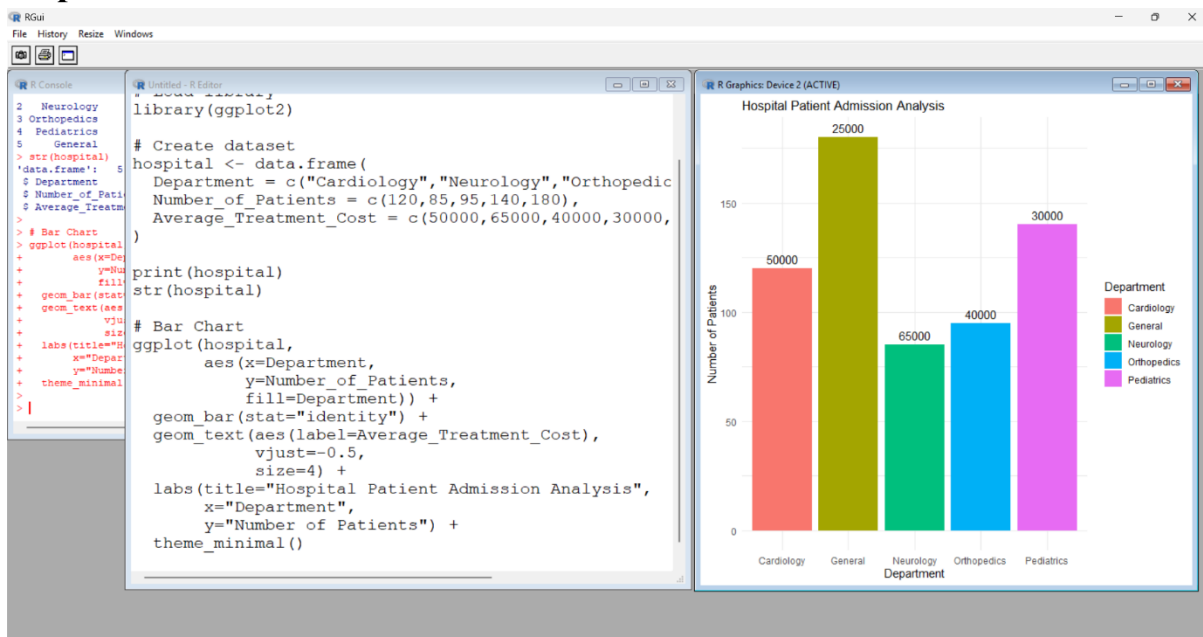
```
      y=Number_of_Patients,
```

```

fill=Department)) +
geom_bar(stat="identity") +
geom_text(aes(label=Average_Treatment_Cost),
          vjust=-0.5,
          size=4) +
labs(title="Hospital Patient Admission Analysis",
      x="Department",
      y="Number of Patients") +
theme_minimal()

```

Output :



4.Weather Monitoring system

Code :

```

install.packages("ggplot2")

library(ggplot2)

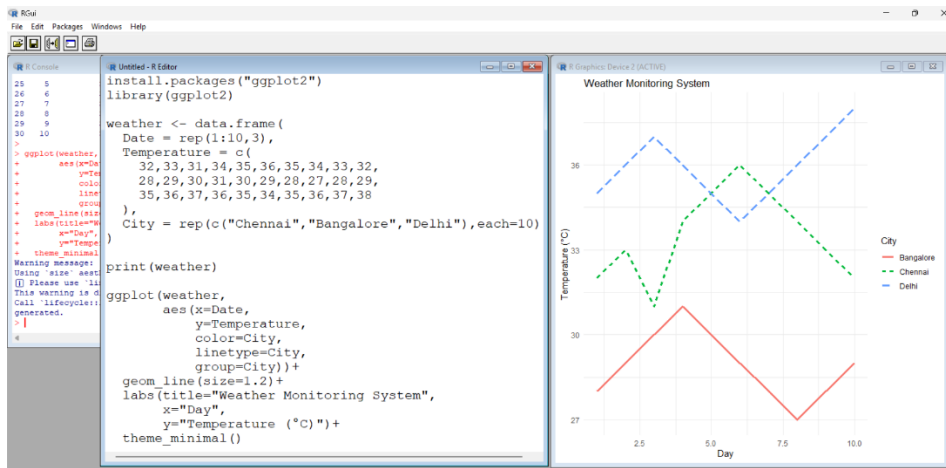
weather <- data.frame(
  Date = rep(1:10,3),

```

```

Temperature = c(
  32,33,31,34,35,36,35,34,33,32,
  28,29,30,31,30,29,28,27,28,29,
  35,36,37,36,35,34,35,36,37,38
),
City = rep(c("Chennai","Bangalore","Delhi"),each=10)
)
print(weather)
ggplot(weather,
  aes(x=Date,
    y=Temperature,
    color=City,
    linetype=City,
    group=City))+
  geom_line(size=1.2)+
  labs(title="Weather Monitoring System",
    x="Day",
    y="Temperature (°C)")+
  theme_minimal()

```



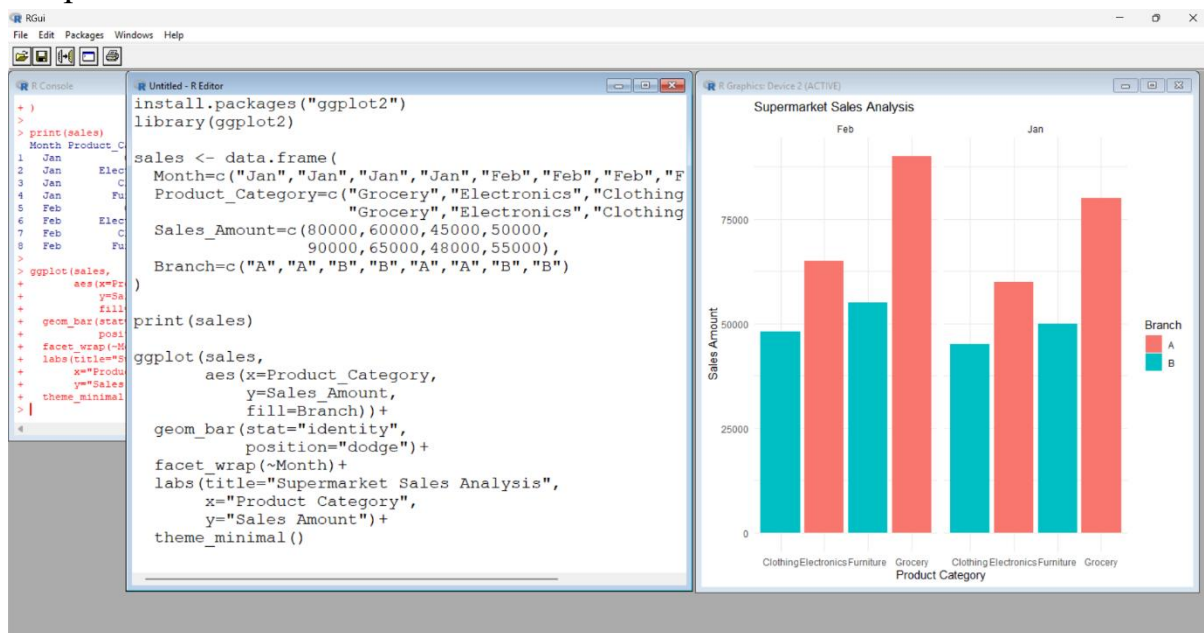
5.Supermarket Sales Analysis :

Code :

```
install.packages("ggplot2")
library(ggplot2)

sales <- data.frame(
  Month=c("Jan","Jan","Jan","Jan","Feb","Feb","Feb","Feb"),
  Product_Category=c("Grocery","Electronics","Clothing","Furniture",
    "Grocery","Electronics","Clothing","Furniture"),
  Sales_Amount=c(80000,60000,45000,50000,
    90000,65000,48000,55000),
  Branch=c("A","A","B","B","A","A","B","B")
)
print(sales)
ggplot(sales,
  aes(x=Product_Category,
    y=Sales_Amount,
    fill=Branch))+
  geom_bar(stat="identity",
    position="dodge")+
  facet_wrap(~Month)+
  labs(title="Supermarket Sales Analysis",
    x="Product Category",
    y="Sales Amount")+
  theme_minimal()
```

Output :



6.Employee salary distribution

Code :

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  Employee_ID=1:20,
```

```
  Salary=c(25000,30000,28000,40000,35000,45000,50000,55000,60000,
```

```
          65000,70000,72000,75000,80000,85000,90000,95000,
```

```
          100000,105000,110000),
```

```
  Experience=c(1,2,2,3,3,4,5,5,6,6,7,8,8,9,10,11,12,13,14,15),
```

```
  Department=c("HR","HR","IT","IT","Sales","Sales","Finance","Finance",
```

```
              "HR","IT","Sales","Finance","HR","IT","Sales",
```

```
              "Finance","HR","IT","Sales","Finance")
```

```
)
```

```
print(employee)
```

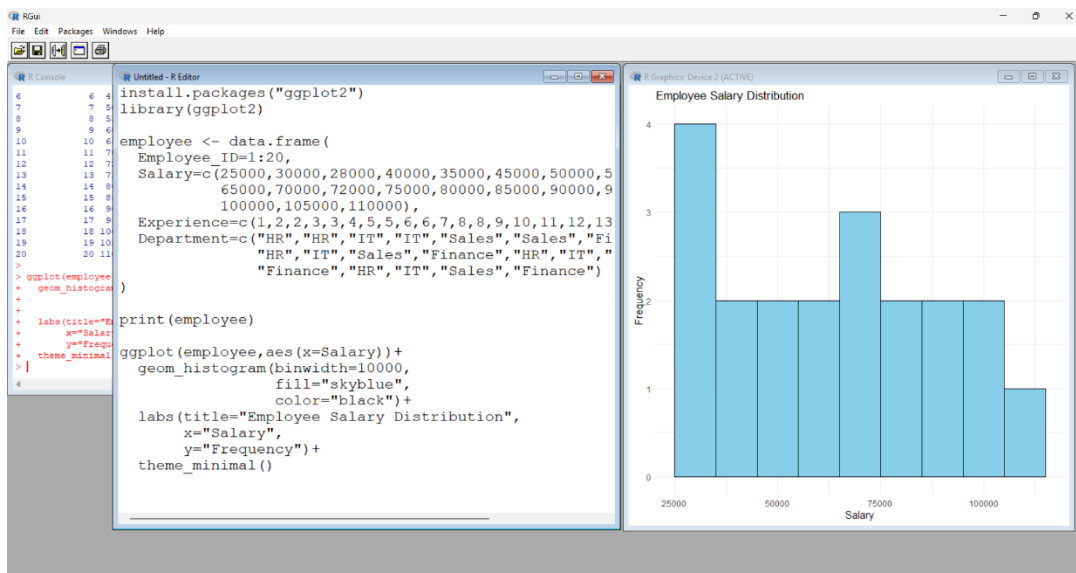
```
ggplot(employee,aes(x=Salary))+
```

```
geom_histogram(binwidth=10000,
               fill="skyblue",
               color="black")+

labs(title="Employee Salary Distribution",
     x="Salary",
     y="Frequency")+

theme_minimal()
```

Output :



7.Smart phone analysis

Code :

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
phone <- data.frame(
```

```
  Brand=c("Samsung","Apple","OnePlus","Xiaomi","Vivo",
          "Oppo","Realme","Motorola"),
```

```
  RAM=c(8,6,12,8,8,12,6,8),
```

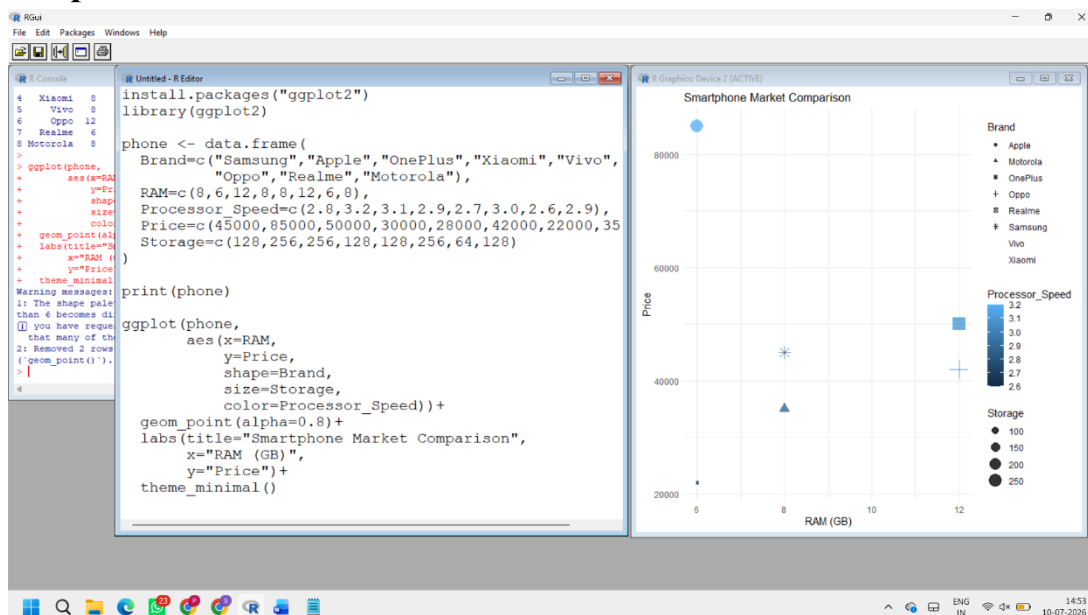
```
  Processor_Speed=c(2.8,3.2,3.1,2.9,2.7,3.0,2.6,2.9),
```

```

Price=c(45000,85000,50000,30000,28000,42000,22000,35000),
Storage=c(128,256,256,128,128,256,64,128)
)
print(phone)
ggplot(phone,
  aes(x=RAM,
      y=Price,
      shape=Brand,
      size=Storage,
      color=Processor_Speed))+
  geom_point(alpha=0.8)+
  labs(title="Smartphone Market Comparison",
       x="RAM (GB)",
       y="Price")+
  theme_minimal()

```

Output :



8.Online Movie Rating Analysis

```

Code : install.packages("ggplot2")
library(ggplot2)

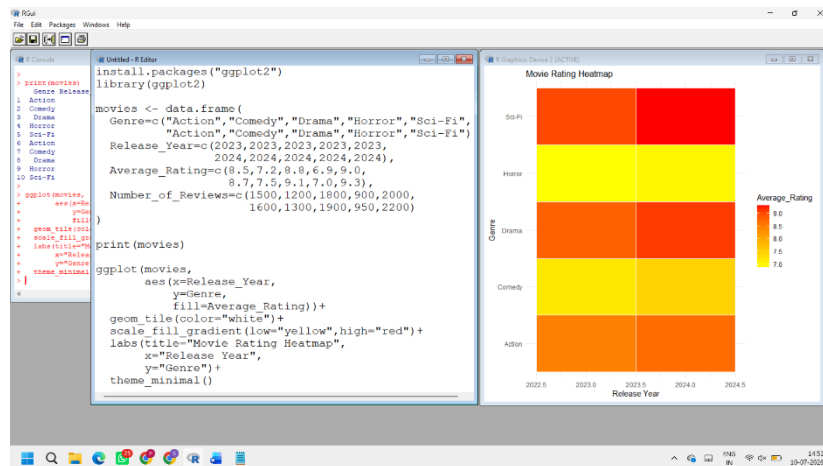
movies <- data.frame(
  Genre=c("Action","Comedy","Drama","Horror","Sci-Fi",
    "Action","Comedy","Drama","Horror","Sci-Fi"),
  Release_Year=c(2023,2023,2023,2023,2023,
    2024,2024,2024,2024,2024),
  Average_Rating=c(8.5,7.2,8.8,6.9,9.0,
    8.7,7.5,9.1,7.0,9.3),
  Number_of_Reviews=c(1500,1200,1800,900,2000,
    1600,1300,1900,950,2200)
)

print(movies)

ggplot(movies,
  aes(x=Release_Year,
    y=Genre,
    fill=Average_Rating))+
  geom_tile(color="white")+
  scale_fill_gradient(low="yellow",high="red")+
  labs(title="Movie Rating Heatmap",
    x="Release Year",
    y="Genre")+ theme_minimal()

```

Output :



9. COVID-19 State-wise Analysis

CODE: `install.packages("ggplot2")`

`library(ggplot2)`

`covid <- data.frame(`

`State=c("Tamil Nadu", "Karnataka", "Kerala", "Maharashtra", "Delhi"),`

`Active_Cases=c(5200, 4300, 6100, 8200, 3000),`

`Vaccination_Percentage=c(92, 90, 94, 91, 89),`

`Population=c(75, 68, 35, 124, 20)`

`)`

`print(covid)`

`ggplot(covid,`

`aes(x=1,`

`y=reorder(State, Active_Cases),`

`fill=Active_Cases))+`

`geom_tile(color="white")+`

`scale_fill_gradient(low="lightyellow",`

`high="darkred"))+`

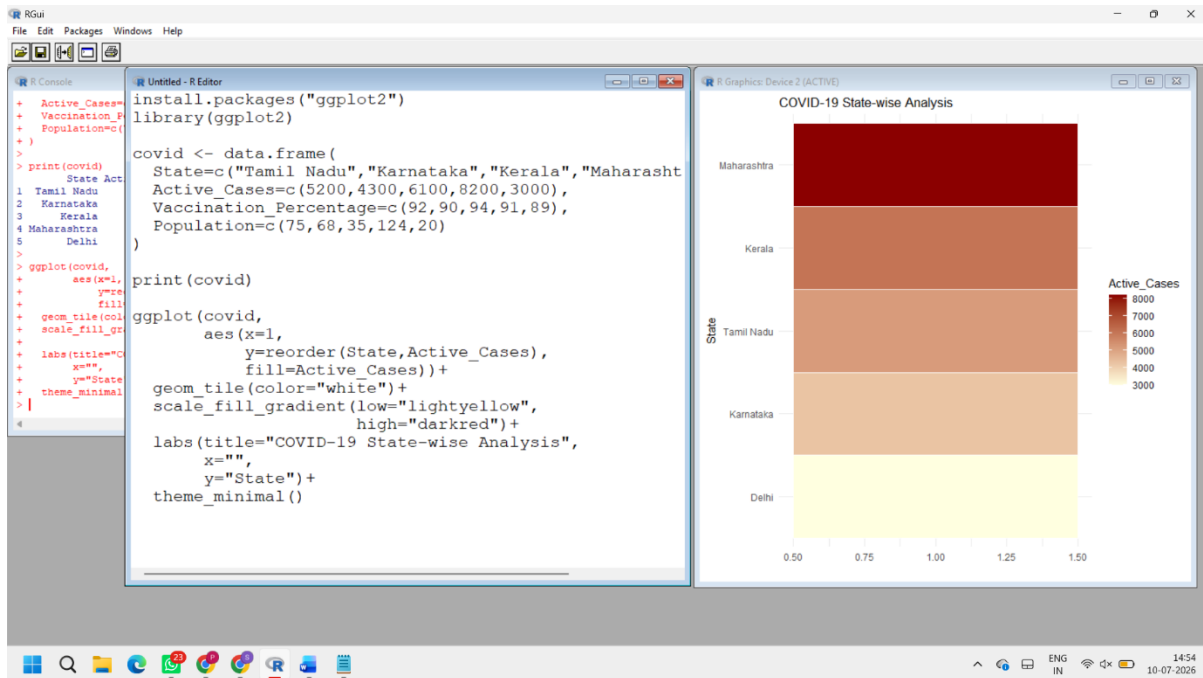
`labs(title="COVID-19 State-wise Analysis",`

`x="",`

`y="State"))+`

```
theme_minimal()
```

OUTPUT:



10. Iris Flower Classification

CODE : `install.packages("ggplot2")`

`library(ggplot2)`

`print(head(iris))`

`ggplot(iris,`

`aes(x=Sepal.Length,`

`y=Petal.Length,`

`color=Species,`

`shape=Species,`

`size=Sepal.Width))+`

`geom_point(alpha=0.8)+`

`labs(title="Iris Flower Classification",`

`x="Sepal Length",`

```
y="Petal Length")+
```

```
theme_minimal()
```

OUTPUT :

